



# MATERIAL SAFETY DATA SHEET

d'Limokleen

## Section 1: PRODUCT AND COMPANY IDENTIFICATION

**Product Name:** d'Limokleen

**Product Number:** 338

**Product Use:** Detergent.

**Manufacturer/Supplier:** Auto Valet Detailing Products  
a div. of 597401 ontario limited  
1366 Sandhill Dr., Ancaster, ON  
L9G 4V5

**Phone Number:** (905) 648-7432

**Emergency Phone:** (905) 648-7832

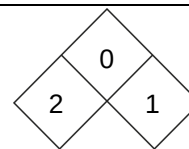
**Date of Preparation:** 1 November 2014

**HMIS Hazard Rating:**

0 = minimal, 1 = slight

2 = moderate, 3 = severe

4 = extreme



Health / Flammability / Physical Hazard

## Section 2: HAZARDS IDENTIFICATION

### EMERGENCY OVERVIEW

**DANGER**

**CORROSIVE MATERIAL – CAUSES BURNS.**

**Potential Health Effects:** See Section 11 for more information.

**Likely Routes of Exposure:** Skin contact, eye contact, inhalation, and ingestion.

**Eye:** Causes burns.

**Skin:** May cause burns. May cause sensitisation by skin contact.

**Ingestion:** May cause burns. May be harmful if swallowed. May cause stomach distress, nausea or vomiting.

**Inhalation:** May cause respiratory tract irritation.

**Chronic Effects:** Prolonged or repeated contact may dry skin and cause irritation.

**Signs and Symptoms:** Symptoms may include discomfort or pain, excess blinking and tear production, with marked redness and swelling of the conjunctiva. Symptoms may include redness, edema, drying, defatting and cracking of the skin.

**Medical Conditions Aggravated By Exposure:** Asthma. Allergies.

**Target Organs:** Skin, eyes, gastrointestinal tract, respiratory system.

**Potential Environmental Effects:** May cause long-term adverse effects in the aquatic environment See Section 12 for more information.

## Section 3: COMPOSITION / INFORMATION ON INGREDIENTS

| Ingredient                              | CAS #      | Wt. %     |
|-----------------------------------------|------------|-----------|
| Monoethanolamine                        | 141-43-5   | 0.5 - 1.5 |
| Tetrapotassium pyrophosphate            | 7320-34-5  | 0.5 - 1.5 |
| Sodium metasilicate                     | 6834-92-0  | 0.5 - 1.5 |
| d-Limonene                              | 5989-27-5  | 1 - 5     |
| Alcohols, C12-14-secondary, ethoxylated | 84133-50-6 | 5 - 10    |
| Dipropylene glycol methyl ether         | 34590-94-8 | 0.5 - 1.5 |
| Ethanol                                 | 64-17-5    | 0.5 - 1.5 |

## Section 4: FIRST AID MEASURES



# MATERIAL SAFETY DATA SHEET

d'Limokleen

- Eye Contact:** In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. If easy to do, remove contact lenses, if worn. Get medical attention immediately.
- Skin Contact:** In case of contact, immediately flush skin with plenty of water. Remove contaminated clothing and shoes. Wash clothing before reuse. Call a physician if irritation develops and persists.
- Inhalation:** If inhaled, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen.
- Ingestion:** If swallowed, do NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person.
- General Advice:** In case of accident or if you feel unwell, seek medical advice immediately (show the label or MSDS where possible).
- Note to Physicians:** Symptoms may not appear immediately.

## Section 5: FIRE FIGHTING MEASURES

**Flammability:** Not flammable by WHMIS criteria.

**Means of Extinction:**

**Suitable Extinguishing Media:** Powder, water spray, foam, carbon dioxide.

**Unsuitable Extinguishing Media:** Not available.

**Products of Combustion:** Oxides of carbon, oxides of nitrogen.

**Explosion Data:**

**Sensitivity to Mechanical Impact:** Not available.

**Sensitivity to Static Discharge:** Not available.

**Protection of Firefighters:** Keep upwind of fire. Wear full fire fighting turn-out gear (full Bunker gear) and respiratory protection (SCBA).

## Section 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions:** Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel.

**Environmental Precautions:** This material is a water pollutant. Keep out of drains, sewers, ditches, and waterways. Minimize use of water to prevent environmental contamination.

**Methods for Containment:** Contain and/or absorb spill with inert material (e.g. sand, vermiculite), then place in a suitable container. Do not flush to sewer or allow to enter waterways. Use appropriate Personal Protective Equipment (PPE).

**Methods for Clean-Up:** Vacuum or sweep material and place in a disposal container. Allow gas to dissipate harmlessly into the atmosphere.

**Other Information:** Not available.

## Section 7: HANDLING AND STORAGE



# MATERIAL SAFETY DATA SHEET

d'Limokleen

## Handling:

Avoid contact with skin and eyes. Do not swallow. Do not breathe gas/fumes/vapor/spray. Handle and open container with care. When using do not eat or drink. Wash hands before eating, drinking, or smoking.

## Storage:

Keep out of the reach of children. Keep container tightly closed. Keep away from food, drink and animal feedingstuffs.

## Section 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

### Exposure Guidelines

| Ingredient                              | Exposure Limits<br>ACGIH-TLV |
|-----------------------------------------|------------------------------|
| Monoethanolamine                        | 3 ppm                        |
| Tetrapotassium pyrophosphate            | Not available.               |
| Sodium metasilicate                     | Not available.               |
| d-Limonene                              | Not available.               |
| Alcohols, C12-14-secondary, ethoxylated | Not available.               |
| Dipropylene glycol methyl ether         | 100 ppm                      |
| Ethanol                                 | 1000 ppm                     |

**Engineering Controls:** Use ventilation adequate to keep exposures (airborne levels of dust, fume, vapor, etc.) below recommended exposure limits.

### Personal Protective Equipment:

**Eye/Face Protection:** Wear eye/face protection.

**Hand Protection:** Wear suitable gloves.

**Skin and Body Protection:** Wear suitable protective clothing.

**Respiratory Protection:** In case of insufficient ventilation, wear suitable respiratory equipment.

**General Hygiene Considerations:** Handle according to established industrial hygiene and safety practices.

## Section 9: PHYSICAL AND CHEMICAL PROPERTIES

|                                  |                  |
|----------------------------------|------------------|
| <b>Appearance:</b>               | Clear.           |
| <b>Color:</b>                    | Orange.          |
| <b>Odour:</b>                    | Citrus.          |
| <b>Odour Threshold:</b>          | Not available.   |
| <b>Physical State:</b>           | Liquid.          |
| <b>pH:</b>                       | 12.8             |
| <b>Viscosity:</b>                | Not available.   |
| <b>Freezing Point:</b>           | ~ 0 °C (~ 32 °F) |
| <b>Boiling Point:</b>            | Not available.   |
| <b>Flash Point:</b>              | Not available.   |
| <b>Evaporation Rate:</b>         | Not available.   |
| <b>Lower Flammability Limit:</b> | Not available.   |



# MATERIAL SAFETY DATA SHEET

d'Limokleen

|                                               |                |
|-----------------------------------------------|----------------|
| <b>Upper Flammability Limit:</b>              | Not available. |
| <b>Vapor Pressure:</b>                        | Not available. |
| <b>Vapor Density:</b>                         | Not available. |
| <b>Specific Gravity:</b>                      | 1.03           |
| <b>Solubility in Water:</b>                   | Complete.      |
| <b>Coefficient of Water/Oil Distribution:</b> | Not available. |
| <b>Auto-ignition Temperature:</b>             | Not available. |
| <b>Percent Volatile, wt. %:</b>               | 84             |
| <b>VOC content, wt. %:</b>                    | Not available. |

## Section 10: STABILITY AND REACTIVITY

**Stability:** Stable under normal storage conditions.

**Conditions of Reactivity:** Heat. Incompatible materials.

**Incompatible Materials:** Oxidizers. Acids. Reducing agents.

**Hazardous Decomposition Products:** Oxides of carbon, oxides of nitrogen.

**Possibility of Hazardous Reactions:** No dangerous reaction known under conditions of normal use.

## Section 11: TOXICOLOGY INFORMATION

### EFFECTS OF ACUTE EXPOSURE

#### Component Analysis

| <b>Ingredient</b>                       | <b>LD<sub>50</sub> (oral)</b> | <b>LC<sub>50</sub></b> |
|-----------------------------------------|-------------------------------|------------------------|
| Monoethanolamine                        | 1720 mg/kg, rat               | Not available.         |
| Tetrapotassium pyrophosphate            | 4640 mg/kg, rat               | Not available.         |
| Sodium metasilicate                     | Not available                 | Not available          |
| d-Limonene                              | 4400 mg/kg, rat               | Not available.         |
| Alcohols, C12-14-secondary, ethoxylated | 3277 mg/kg oral, rat          | Not available.         |
| Dipropylene glycol methyl ether         | 5400 uL/kg, rat               | Not available.         |
| Ethanol                                 | 7060 mg/kg, rat               | 20000 ppm 10hrs,rat    |

**Eye:** Causes burns. Symptoms may include discomfort or pain, excess blinking and tear production, with marked redness and swelling of the conjunctiva.

**Skin:** Causes burns. May cause sensitisation by skin contact. Symptoms may include redness, edema, drying, defatting and cracking of the skin.

**Ingestion:** Causes burns. May be harmful if swallowed. May cause stomach distress, nausea or vomiting.

**Inhalation:** May cause respiratory tract irritation.

### EFFECTS OF CHRONIC EXPOSURE

**Target Organs:** Not available.

**Chronic Effects:** Not hazardous by WHMIS criteria.

**Carcinogenicity:** Not hazardous by WHMIS criteria.

| <b>Ingredient</b> | <b>Chemical Listed as Carcinogen or Potential Carcinogen *</b> |
|-------------------|----------------------------------------------------------------|
|-------------------|----------------------------------------------------------------|



DETAILING PRODUCTS

## MATERIAL SAFETY DATA SHEET

d'Limokleen

|                                         |             |
|-----------------------------------------|-------------|
| Monoethanolamine                        | Not listed. |
| Tetrapotassium pyrophosphate            | Not listed. |
| Sodium metasilicate                     | Not listed. |
| d-Limonene                              | Not listed. |
| Alcohols, C12-14-secondary, ethoxylated | Not listed. |
| Dipropylene glycol methyl ether         | Not listed. |
| Ethanol                                 | Not listed. |

\* See Section 15 for more information.

**Mutagenicity:** - MAK-5, minimal risk at usual workplace concentrations

**Reproductive Effects:** Not hazardous by WHMIS criteria.

**Developmental Effects:**

**Teratogenicity:** Not hazardous by WHMIS criteria.

**Embryotoxicity:** Not hazardous by WHMIS criteria.

**Respiratory Sensitization:** Not hazardous by WHMIS criteria.

**Skin Sensitization:** Hazardous by WHMIS criteria.

**Toxicologically Synergistic Materials:** Not available.

### Section 12: ECOLOGICAL INFORMATION

**Ecotoxicity:** May cause long-term adverse effects in the aquatic environment

**Persistence / Degradability:** Not available.

**Bioaccumulation / Accumulation:** Not available.

**Mobility in Environment:** Not available.

### Section 13: DISPOSAL CONSIDERATIONS

**Disposal Instructions:**

This material and its container must be disposed of in a safe way.

### Section 14: TRANSPORTATION INFORMATION

**TDG Classification**

Not regulated

### Section 15: REGULATORY INFORMATION

**Federal Regulations**

**Canadian:** This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations and the MSDS contains all the information required by the Controlled Products Regulations.

**Global Inventories**

**Ingredient**

**Canada**



# MATERIAL SAFETY DATA SHEET

d'Limokleen

Monoethanolamine  
Tetrapotassium pyrophosphate  
Sodium metasilicate  
d-Limonene  
Alcohols, C12-14-secondary, ethoxylated  
Dipropylene glycol methyl ether  
Ethanol

## DSL/NDSL

DSL  
DSL  
DSL  
DSL  
DSL  
DSL  
DSL

### WHMIS Classification(s):

Class D2A – MAK-5 minimal mutagenic risk in industrial workplace.  
Class D2B - Skin Sensitization  
Class E - Corrosive Material

### WHMIS Hazard Symbols:



### SOURCE AGENCY CARCINOGEN CLASSIFICATIONS:

**OSHA (O)** Occupational Safety and Health Administration.

**ACGIH (G)** American Conference of Governmental Industrial Hygienists.

- A1 - Confirmed human carcinogen.
- A2 - Suspected human carcinogen.
- A3 - Animal carcinogen.
- A4 - Not classifiable as a human carcinogen.
- A5 - Not suspected as a human carcinogen.

**IARC (I)** International Agency for Research on Cancer.

- 1 - The agent (mixture) is carcinogenic to humans.
- 2A - The agent (mixture) is probably carcinogenic to humans; there is limited evidence of carcinogenicity in humans and sufficient evidence of carcinogenicity in experimental animals.
- 2B - The agent (mixture) is possibly carcinogenic to humans; there is limited evidence of carcinogenicity in humans in the absence of sufficient evidence of carcinogenicity in experimental animals.
- 3 - The agent (mixture, exposure circumstance) is not classifiable as to its carcinogenicity to humans.
- 4 - The agent (mixture, exposure circumstance) is probably not carcinogenic to humans.

**NTP (N)** National Toxicology Program.

- 1 - Known to be carcinogens.
- 2 - Reasonably anticipated to be carcinogens.

## Section 16: OTHER INFORMATION

### Disclaimer:

The information contained in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. It is the user's responsibility to satisfy oneself as to the suitability and completeness of this information for the user's own particular use.

**Expiry Date:** 1 November 2017

**Version #:** 1.0

**Prepared by:** H & S Department  
Phone: (905) 648-7832